

BREAK INTO AEROSPACE

Aircraft Load Controller

Career & Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: The aircraft load controller produces one document that legally authorises every commercial flight to depart. The load sheet — signed by the load controller and transmitted to the aircraft's captain — certifies the aircraft's take-off weight, zero fuel weight, landing weight, and centre of gravity position, and confirms that the load has been distributed across the holds in accordance with the structural limits of the aircraft and the approved loading instructions. Without it, the flight does not go. With an error in it, the flight departs with incorrect performance assumptions, potentially incorrect trim settings, and a weight or CG that the captain believes to be safe but is not.

This is not a role that sounds dramatic, but its safety consequences are direct and documented. The Air Midwest Flight 5481 crash in Charlotte in 2003 was partially attributable to an aft CG that caused the aircraft to pitch up uncontrollably during takeoff rotation. The Moorea Blue Airlines ATR 72 incident and multiple runway exceedances in the accident record have load control errors as contributing factors. Every load sheet that leaves the dispatch desk with an error in it is a potential entry into that record, and every load controller who understands this — not as a policy statement but as a daily professional reality — is doing their job correctly.

The role sits at the intersection of aviation operations, aircraft performance engineering, dangerous goods regulation, and real-time logistics management. On a busy short-haul morning bank at a major European hub, a load controller may issue thirty or forty load sheets in four hours, each requiring accurate take-off weight calculation, CG check, hold loading verification against structural limits, dangerous goods confirmation, and ACARS transmission to the aircraft — all while managing late bags, last-minute cargo changes, fuel uplifts that differ from the planned figure, and captains calling from the cockpit asking for an update on the zero fuel weight.

The employer landscape is concentrated among the major ground handling agents who hold the load control contract at most airports: Swissport, Menzies Aviation (formerly Airline Services Group), dnata, WFS (Worldwide Flight Services), and Aviapartner. Some major network carriers maintain in-house load control at their hubs — Lufthansa at Frankfurt and Munich, British Airways at Heathrow, Air France at CDG, KLM at Schiphol, Turkish Airlines at Istanbul. Low-cost carriers typically outsource load control to ground handlers. Cargo carriers — DHL Aviation, FedEx Express, UPS Airlines, Cargolux, Air Hong Kong — employ load controllers with specific cargo loading expertise and dangerous goods knowledge at a deeper level than the passenger carrier equivalent.

PART I: WHAT THE FIELD ACTUALLY LOOKS FOR

By Level

Trainee Load Controller

Entry to load control requires no prior aviation experience in most organisations — the technical knowledge is trained in. What employers are assessing at the trainee level is the combination of numeracy, attention to detail, and the ability to work accurately under time pressure. The selection process for ground handler load control trainees typically includes a numerical reasoning test (not a maths exam — a test of how quickly and accurately someone can handle numbers under time pressure), sometimes a basic logical reasoning test, and an interview that probes customer service orientation and stress tolerance alongside the safety attitude

questions that every aviation employer asks.

The specific character trait that load control employers look for — and that distinguishes candidates who will succeed from those who will not — is the willingness to stop the clock when something does not look right, even when the aircraft is late and the operations controller is pressing for the load sheet. This is not stubbornness — it is the professional discipline of someone who understands that their signature on the load sheet represents a personal attestation that the data is correct. Candidates who have worked in any environment requiring this kind of accuracy under operational pressure — pharmacy dispensing, laboratory testing, financial reconciliation, surgical instrument counting in a medical context — are demonstrating the character the role requires.

Certified Load Controller (0–3 years)

After completing the initial training course and qualifying on the first aircraft type, the load controller is certificated to issue load sheets independently for that type. The first-year interview for a role at a new employer focuses on the candidate's type qualifications, their familiarity with the specific load systems used at the operation (Altea, UFIS, Kestrel, or the airline-specific system used by the contracting carrier), their understanding of the IATA Loading and Unloading of Aircraft procedures, and their approach to the specific safety scenarios that every load control interview includes: "the aircraft has a last-minute upgrade of a passenger from economy to business class — the seat change has been processed in the departure control system but you are about to issue the load sheet. What do you check?" The expected answer considers the CG implication of the seat zone change, checks that the passenger has been repositioned in the weight and balance system, confirms the final passenger count and distribution, and only then issues the load sheet.

Senior Load Controller / Load Control Supervisor (3–8 years)

At this level, the interview shifts to quality oversight and exception management. The senior load controller is responsible not just for their own load sheets but for the quality of the team's output — for reviewing and signing off complex or unusual loads, for making the call when a last-minute change is too significant to accommodate within the available time, and for managing the relationship with the airline's performance team when a situation falls outside the standard loading instructions. The stories that reveal senior-level competence are the ones where the load controller identified an error — in their own work, in a colleague's work, or in the data coming from the departure control system — before the flight departed, and can describe the process they followed to correct it and the documentation they created.

Load Control Trainer / Quality Auditor

This pathway moves from operational load control into the training and quality infrastructure of the operation. Load control trainers write and deliver the aircraft type qualification courses that certify load controllers for specific aircraft. Quality auditors conduct the proficiency checks required every 12–24 months under most airlines' approved training organisations. These roles require deep technical knowledge of the aircraft loading documents — the Weight and Balance Manual, the Loading Instructions Reference Document, the Aircraft Loading Manual — as well as the adult learning methodology needed to convey that knowledge to trainees who are encountering the aircraft type for the first time.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

- **CG calculation from first principles** — the index system and the trim sheet. The aircraft's centre of gravity position determines how much pitch trim the flight crew must set before takeoff, and whether the aircraft will be stable throughout the flight as fuel is burned and passengers move. Modern load systems calculate CG automatically, but understanding the calculation from first principles — the index system that expresses the moment of each loaded zone (holds, cabin zones, fuel tanks) as an index value relative to the datum, the summation of index values to find the loaded CG index, and the conversion of index to CG as a percentage of Mean Aerodynamic Chord (%MAC) — is the technical foundation that distinguishes load controllers who can diagnose a system error or an unusual loading situation from those who only know how to operate the software. Every serious load control interview at senior level includes a manual CG calculation exercise, and candidates who have practised this for the aircraft types they hold qualifications on are immediately distinguishable.
- **The Weight and Balance Manual** — structure, limitations, and how to navigate it. The Weight and Balance Manual (WBM, sometimes called the Aircraft Weight and Balance Manual or AWB) is the primary technical document governing the weight and balance of a specific aircraft type. It contains the Maximum Structural Take-Off Weight (MSTOW), the Maximum Structural Landing Weight (MSLW),

the Maximum Structural Zero Fuel Weight (MSZFW), the forward and aft CG limits as a function of weight, the hold structural limits (the maximum weight per running metre of floor loading and the maximum total hold weight), the passenger weight standards, the fuel density and fuel index data, and the operational procedures for unusual loading situations. Understanding how to navigate the WBM — not just to find a specific limit but to interpret the CG envelope diagram, read the CG limit curves, and understand why the forward CG limit becomes more restrictive at low weights (tail-heaviness for pitch control) and the aft limit becomes more restrictive at high weights (stability requirement) — is the core technical literacy of the load control role.

- **IATA Dangerous Goods Regulations** — categories, packaging, labelling, and the load controller's specific obligations. The load controller is a critical point in the dangerous goods chain. Their obligation is not to classify or package dangerous goods — that is the shipper's and the cargo acceptance team's responsibility — but to verify that the dangerous goods information on the Notification to Captain (NOTOC) is consistent with the aircraft's limitations, that dangerous goods are not loaded in positions that would violate the segregation requirements (certain dangerous goods classes must be separated from food, from animals, from people), and that the captain has been informed of the dangerous goods on board through the NOTOC before departure. Specifically, the load controller must know the categories of dangerous goods that are forbidden on passenger aircraft (certain divisions of Class 1, Class 6.2 infectious substances, Class 7 radioactive materials above the threshold level) and must recognise when a NOTOC contains an item that should not be on the flight. IATA DGR Category 6 (loading) and Category 8 (passenger and crew) are the specific DGR training categories required for load control staff.
- **Hold loading restrictions** — floor load limits, per-bay limits, and aircraft-specific structural constraints. Every aircraft hold has structural limitations beyond the total maximum hold weight: a maximum weight per running metre of floor loading, maximum per-bay weights, restrictions on the stacking of certain cargo types, and incompatibility rules for certain dangerous goods and cargo categories. Understanding how to calculate the floor load of a specific piece of cargo — the total weight divided by the base area in contact with the floor, compared against the kg/m² floor load limit from the Aircraft Loading Manual — and how to distribute cargo within a hold to stay within the per-bay limits while optimising the CG is the practical loading knowledge that experienced load controllers use when loading an unusual consignment. The specific floor load calculation is tested in load control type qualification training and is a common technical question in interviews for cargo operations roles.
- **Passenger and cargo weight standards** — IATA standard weights, airline-specific weights, and the seasonal adjustment. Aircraft weight and balance calculations use standard passenger weights because weighing individual passengers is impractical. IATA Resolution 700 provides standard passenger weights that airlines may use: 84 kg for male, 70 kg for female, and 35 kg for children — these are the standard values used as the default in the absence of airline-specific data. Many airlines conduct their own passenger weight surveys and use airline-specific standard passenger weights that may differ from IATA standards. The load controller must know which standard is applicable to the operation and must ensure the correct standard is applied in the load system, because using the wrong passenger weight standard shifts the calculated take-off weight and CG by potentially safety-significant amounts for a full aircraft.
- **The departure control system** — Altea, Altéa DCM, UFIS, Kestrel, Radixx, or carrier-specific systems. The departure control system (DCS) is the software platform that manages the passenger check-in, boarding, and load control process. The load control module of the DCS — typically Amadeus Altea DCM, SITA UFIS, or a carrier-proprietary system — generates the load sheet and trim sheet from the passenger, cargo, fuel, and mail data recorded in the system. Understanding how the DCS captures passenger data (check-in weights, seat assignments, transit passengers), how it handles last-minute changes (LMC — last minute change — procedures), and how the load control module uses the data to calculate the load sheet are the operational system competencies that every experienced load controller has and that interviews for roles at specific carriers or handlers probe through scenario questions based on the system that operation uses.
- **Fuel planning interface** — the relationship between fuel uplift, zero fuel weight, and take-off weight. The load controller does not plan the fuel — that is the flight dispatcher or pilot's responsibility — but

must incorporate the finalised fuel figure into the load sheet calculation. Understanding how the fuel figure affects the take-off weight ($MTOW = ZFW + \text{fuel}$, and $MTOW$ must not exceed the structural $MTOW$ or the performance-limited TOW calculated by the performance team), how the fuel distribution across tanks affects the CG (a fuel load in the centre tank moves the CG differently from the same fuel distributed equally between wing tanks), and how a last-minute fuel change propagates through the load sheet (requiring recalculation of TOW , CG , and possibly a recheck of the performance-limited TOW) is the fuel system interface literacy that every load controller requires.

- **Loading Instructions Release Document (LIRD)** — what it is, where it comes from, and how it governs the physical loading. The Loading Instructions Release Document (sometimes called the Load Plan or the Loading Instruction) is the document that the load controller issues to the ramp agent who physically loads the aircraft. It specifies which items go in which hold position, the load sequence (items loaded last come off first — LIFO for transit cargo, specific sequence for connecting bags), any hold-specific restrictions, and the verification that the ramp agent must make before closing and signing the hold document. Understanding how the LIRD is generated from the load plan, what the ramp agent's obligation is when they receive it (to load according to the instruction, to report any discrepancy between the instruction and the actual cargo presented, and to return a signed copy confirming completion), and what the load controller must do if the LIRD cannot be followed exactly (reassess the CG impact, obtain authority from the airline if required, reissue the load sheet) is the loading documentation literacy that every load control interview probes.
- **ACARS and datalink** — how load data is transmitted to the aircraft. Most commercial aircraft receive the load sheet and performance data electronically through ACARS (Aircraft Communications Addressing and Reporting System) rather than by paper delivery. The load controller transmits the finalised load sheet through the ACARS ground station, and the aircraft's flight management system or a dedicated load sheet printer in the cockpit produces the document for the captain to review and sign. Understanding the ACARS transmission process — how to initiate the transmission from the load system, how to confirm receipt and acknowledgement by the aircraft, and what the procedure is if ACARS transmission fails and a manual or verbal load sheet delivery is required — is the operational competency that every load controller working on aircraft with ACARS capability must have.

How Load Control Interviews Differ from Generic Aviation Operations Roles

- **The safety scenario question is not a formality — it is the primary assessment.** Every load control interview, at every level and every employer, includes one or more safety scenario questions. The scenarios are not hypothetical academic exercises — they are realistic operational situations that load controllers encounter regularly: the passenger manifest shows 183 passengers but the boarding gate reported 181 — what do you do before issuing the load sheet? The ramp agent calls to say the cargo in position 4 of hold 1 has shifted during the push-off the stand — what is the procedure? The captain calls from the cockpit to say the trim setting does not match the value on the load sheet — what do you do? The expected answers to these scenarios are specific, regulatory, and procedural — not vague expressions of good judgment. Candidates who know the specific answer (stop the departure, reconcile the discrepancy, do not issue or re-issue the load sheet until the discrepancy is resolved) are demonstrating genuine operational knowledge; candidates who give generic "I would investigate the issue" answers are revealing they have studied the concept but not the procedure.
- **Numerical accuracy under time pressure is specifically tested.** Many load control employers include a numerical test in their selection process — sometimes a dedicated psychometric numerical reasoning test, sometimes a simplified manual load sheet exercise — because the ability to work with numbers quickly and accurately under time pressure is the primary job competency. Candidates who have not practised this specific type of numerical work before the interview will underperform relative to their true capability. The specific numbers involved — weights in kilograms to the nearest kilogram, moments or indices to two decimal places, CG as a percentage of MAC to one decimal place — are not conceptually difficult, but the combination of precision requirement, time pressure, and operational context creates a specific performance challenge that benefits from preparation.
- **Type-specific knowledge is tested at operational depth.** "Tell me about the A320's weight limits" is not the question. "The load sheet for the flight shows 73,400 kg take-off weight and the captain has called to

say the Zero Fuel Weight is 2,000 kg higher than planned because of a cargo substitution — is the flight legal, and what are the first three things you check?” is the question. The expected answer references the MSZFW and MSTOW for the specific A320 variant, the performance-limited TOW that the performance team has issued for that departure, and the CG implications of the cargo substitution. This is operational knowledge that comes from having worked on that aircraft type, not from having read about it.

- **The relationship with other operational teams is assessed.** Load control sits at the centre of a network of operational teams: the ramp, the departure gate, the cargo acceptance team, the fuel dispatcher, the flight crew, the airline’s operations control, and the performance engineering team. Candidates who understand the information flows between these teams — who knows when to call the performance team for a new take-off weight limit versus when the existing limit still applies, when a change is significant enough to require captain notification versus when it falls within the standard LMC procedure, and when the load controller must refuse to issue the load sheet until the authority team has been contacted — are demonstrating the operational integration knowledge that supervisors and trainers look for.
- **The just culture and error reporting expectation is tested explicitly.** Load control is a safety-critical role with a paper trail — every load sheet is archived, every ACARS transmission is logged, and every error has a potential path to an incident report. Load control employers specifically probe whether candidates understand that discovering an error — in their own work or a colleague’s — must result in notification and correction before the flight departs, regardless of the time pressure. The candidate who describes a scenario where they corrected a colleague’s error quietly without notifying the supervisor is revealing a quality culture gap that experienced load control managers find as concerning as the error itself.

PART II: 20 SPECIFIC PREP TASKS

TRAINEE / ENTRY LEVEL

1. Understand what the load sheet is, what it authorises, and why errors have direct safety consequences 3–4 hours

Skill: The fundamental professional context of the role — understanding what you are actually signing

Why the field cares: The load sheet is the document that legally authorises the flight to depart at the stated weight and CG. The captain accepts responsibility for the aircraft by signing or acknowledging the load sheet. If the load sheet contains an error, the captain is operating the aircraft based on incorrect information — incorrect take-off weight means the performance team’s V-speeds and runway requirement calculations are wrong; incorrect CG means the trim setting is wrong; incorrect hold loading means the structural loading of the floor may be exceeded. Candidates who understand this chain of consequence — not as a regulation to be memorised but as a direct physical and legal accountability — are demonstrating the professional attitude that load control requires. Every interview begins with this understanding as a baseline check.

Done signal: You can explain, in your own words, what the load sheet contains (MTOW, ZFW, MLW, CG, fuel at takeoff, distribution of load by hold and cabin zone), what it authorises (the captain to depart with the stated load), why it must be accurate (the captain’s performance calculations, trim setting, and structural safety depend on it), and what the consequence of a specific error type would be for the flight (an understatement of ZFW inflates the TOW margin, an aft CG error causes incorrect trim). You can name one real accident in which load control or weight and balance was a contributing factor.

2. Learn to perform a manual CG calculation using the index method — without a computer 6–8 hours

Skill: The foundational weight and balance calculation that every serious load control interview tests

Why the field cares: Modern load systems calculate CG automatically, but the load controller who

6–8 hours

cannot perform the calculation manually cannot verify the system's output or diagnose a discrepancy between the computed CG and the expected CG. Every experienced load controller knows how the index system works: each loading zone (forward hold, aft hold, business class, economy rows 1–20, rows 21–40, fuel in wing tanks, fuel in centre tank) has an index arm value that represents its moment relative to the aircraft datum; the total moment index is the sum of the products of each load weight and its index arm; and the loaded CG is derived from the total moment index divided by the total weight. The conversion from this index value to CG as a percentage of MAC requires the aircraft-specific conversion table from the WBM. Candidates who have practised this calculation on a specific aircraft type — even on a simplified version using publicly available WBM excerpts for training purposes — are demonstrating the mathematical engagement with the role that generic numerical aptitude tests cannot reveal.

Done signal: You can work through a complete manual weight and balance calculation for a simplified aircraft scenario — given a basic weight, a fuel weight and its index contribution, passenger counts in two or three cabin zones with their index arms, and cargo weights in two hold positions with their index arms — to arrive at the take-off weight, zero fuel weight, and CG index. You can read the CG index off the trim sheet or CG envelope diagram and determine whether the result is within the forward and aft limits for the take-off weight. You have practised this for a specific aircraft type using its actual WBM data.

3–4 hours

3. Understand the Maximum Structural Weights — MTOW, MZFW, MLW, and what limits each

Skill: The weight limits that govern every load sheet issued

Why the field cares: Every load sheet calculation is bounded by four maximum structural weights: the Maximum Structural Take-Off Weight (MSTOW), the Maximum Structural Zero Fuel Weight (MSZFW), the Maximum Structural Landing Weight (MSLW), and the Maximum Payload Weight. Understanding not just the values for a specific aircraft type but what each weight represents physically — why the MSZFW is a wing bending load limit (the wing root bending moment is maximum at MSZFW when the wing fuel tanks are empty and all weight is in the fuselage), why the MSLW is a landing gear structural limit (the energy absorbed in a hard landing is proportional to the landing weight), and why performance-limited take-off weights are always less than or equal to MSTOW (never greater) — is the weight limits literacy that distinguishes a trained load controller from a candidate who has read the glossary.

Done signal: You can state, for the aircraft type you are preparing for, the MSTOW, MSZFW, MSLW, and explain what structural consideration limits each. You can explain why the zero fuel weight limit exists and why its violation is serious (the wing bending moment). You can explain how the performance-limited TOW interacts with the structural limits — specifically that the load sheet must comply with both, and that the governing limit is whichever is lower for the specific departure.

2–3 hours

4. Prepare behavioural answers connecting accuracy and communication to specific safety outcomes

Skill: STAR stories with the stopped-the-departure-to-get-it-right emphasis load control employers probe

Why the field cares: Load control interviews consistently probe situations where the candidate was under time pressure but had to stop, check, or communicate something before proceeding. The stories that are weakest describe situations where the candidate did their job without incident. The strongest describe situations where something was wrong — a weight discrepancy, a NOTOC that didn't match the manifest, a gate agent reporting a different passenger count than the system showed — and the candidate identified it, stopped the process, resolved the discrepancy, and documented the resolution before issuing the load sheet. These stories must be genuine; fabricated stories collapse under specific follow-up questions from experienced load control supervisors who know every common scenario.

Done signal: Four STAR stories covering: a situation where you identified a numerical discrepancy

in data you were responsible for before it caused a problem; a time you communicated an error or a concern upward to a supervisor despite time pressure; a situation where you worked accurately under time pressure and specific steps you took to maintain accuracy; and a time you had to ask for help or more time to complete a task correctly rather than rushing. These stories do not have to be from aviation — they may be from any environment requiring accuracy under operational pressure.

2–3 hours

5. Understand the Minimum Equipment List interface for load control — how MEL items affect the load sheet

2–3 hours

Skill: The operational connection between aircraft technical status and load control limitations

Why the field cares: Some MEL items have direct implications for load control. A hold cargo door that has a deferred defect and is limited to a reduced maximum opening size may restrict the maximum cargo item dimension that can be loaded. An aft hold heating system on MEL may prohibit the carriage of live animals in that hold. A fuel quantity indicating system fault may require a manual fuel figure that the dispatcher must confirm before the load sheet is issued. A trim indicator system fault may require the load controller to calculate and communicate the trim setting to the flight crew rather than relying on the automatic computation. Load controllers who understand how to check the aircraft technical log for MEL items that affect loading, and who know the standard MEL-load control interface procedure at their operator, are demonstrating the operational completeness that experienced supervisors require.

Done signal: You can explain the general principle of how MEL items interact with load control — that a MEL item affecting a hold system, a cargo loading system, or a fuel system may create additional loading restrictions that must be checked before the load sheet is issued. You can describe the standard process for checking whether any active MEL items affect the loading for a specific departure, and explain what the load controller does when they identify a MEL-related loading restriction that was not flagged in the original load plan.

6. Understand IATA Dangerous Goods Regulations at the load controller level — Categories 6 and 8

4–5 hours

Skill: The DGR knowledge that is a regulatory requirement for load control certification

Why the field cares: IATA DGR Category 6 (Loading of Dangerous Goods) and Category 8 (Dangerous Goods in Passenger Baggage) are the specific training categories required for load control staff. Category 6 covers the load controller's obligations regarding dangerous goods: checking the NO-TOC against the aircraft's limitations for the dangerous goods category and quantity, verifying that segregation requirements are reflected in the load plan, and ensuring the captain receives the NO-TOC before departure. Category 8 covers the restrictions on dangerous goods that passengers may carry — the items that are permitted in carry-on or checked baggage under the allowed quantities, and the items that are prohibited entirely. Understanding both is a regulatory requirement for load control qualification at any IATA-member carrier.

Done signal: You can describe the load controller's specific obligations when a NOTOC is presented — what you check against the aircraft's dangerous goods limitations, how you verify segregation requirements in the load plan, and how you confirm captain notification. You can name the classes of dangerous goods that are forbidden in passenger aircraft holds. You can describe what to do if a NO-TOC is presented with an item that you believe is not permitted on the flight. You have completed or are enrolled in the relevant IATA DGR training.

CERTIFIED LOAD CONTROLLER (0–3 years)

All trainee tasks still apply. Behavioural stories must come from real operational contexts. Type-specific knowledge must be for the aircraft types the employer operates.

7. Build thorough working knowledge of the Last Minute Change procedure — what triggers it, what is required, and who must be notified

3–4 hours

Skill: The procedure governing late changes to a finalised load sheet — the most common source of load control errors

Why the field cares: Last Minute Changes — changes to the load that occur after the load sheet has been issued — are the most operationally complex and error-prone part of load control. A LMC may be triggered by a late-checking passenger, a passenger no-show, a cargo item arriving late, or a hold that could not be loaded as planned. IATA procedures define what weight and CG change is permissible without reissuing the full load sheet — typically a small weight change within a defined delta that does not affect the CG by more than an established tolerance. Changes beyond this tolerance require a new load sheet to be calculated, issued to the captain, and acknowledged before departure. The load controller who processes a LMC correctly — recording the change, recalculating the weight and CG impact, determining whether the change is within the LMC tolerance or requires a new load sheet, notifying the captain appropriately, and documenting everything — is doing load control correctly. The load controller who accepts a LMC verbally and does not document it is creating a safety gap.

Done signal: You can describe the LMC procedure for the operation you work in: what change triggers an LMC versus a new load sheet, what the weight delta tolerance is for a LMC (typically ± 500 kg or $\pm 1,000$ kg depending on the airline procedure), what documentation must be created for every LMC, and what the captain notification procedure is for a LMC versus a new load sheet. You can describe a specific LMC scenario — a late-checking passenger who weighs significantly more than the standard passenger weight — and walk through the calculation and procedure from first principles.

8. Understand hold loading — per-bay limits, floor load calculation, and the distribution requirements for standard unit load devices

3–4 hours

Skill: The physical loading constraints that govern what goes where in the aircraft hold

Why the field cares: Aircraft holds have structural loading limits that go beyond the total maximum hold weight. Each hold bay has a maximum weight, and the floor has a maximum loading in kilograms per square metre or kilograms per running metre. For standard unit load devices (ULDs — LD3, LD6, LD11, LD26, and equivalents), the positions are standardised and the load system manages compliance automatically. For bulk loading — individual pieces of cargo placed directly on the hold floor rather than in a ULD — the load controller must verify that the weight distribution does not exceed the floor load limits. Understanding how to calculate the floor load for a specific cargo item, where to find the floor load limits in the Aircraft Loading Manual, and what the procedure is when a cargo item is too heavy for the available floor loading are the bulk loading competencies that cargo-heavy operations specifically probe.

Done signal: You can calculate the floor load in kg/m^2 for a specified cargo item — given its weight and base dimensions — and compare the result against the floor load limit from the Aircraft Loading Manual for the relevant hold position. You can describe what the load controller's obligation is if the calculated floor load exceeds the limit (the item cannot be loaded in that position; either a different position must be found, a spread-load platform must be used, or the item must be removed from the flight). You can describe the standard ULD types and their typical tare weights and volume capacities.

9. Prepare a story about identifying and resolving a load sheet discrepancy before departure

2 hours

Skill: Applied load control accuracy at the level experienced supervisors specifically probe

Why the field cares: The most revealing interview question in load control is “tell me about a time you found an error in a load sheet — yours or someone else's — before the aircraft departed, and what you did.” Every experienced load controller has a story of this type — a passenger manifest that was short by two, a hold weight that did not match the cargo receipt, a fuel figure that had not been updated after a late uplifted change. The story reveals whether the candidate's response was

2 hours

procedurally correct (stop, verify, correct, document, notify) or shortcut (update the figure without a formal LMC, ask the gate agent to “just add it to the manifest,” or let the small discrepancy go because the flight was late). The procedurally correct response, even when it caused a delay, is the right story to tell.

Done signal: You have a genuine story about a load sheet discrepancy — either discovered in your own work before transmission, in a colleague’s work, or identified by the captain or ramp agent during the pre-departure check. The story describes the specific discrepancy, the action you took (stop, verify against the source data, correct the calculation, issue a new load sheet or LMC as appropriate, notify relevant parties, document), and the outcome. The story demonstrates procedural compliance under time pressure and a willingness to cause a delay rather than issue an incorrect load sheet.

4–6 hours

10. Build working knowledge of departure control system operation — Altea, UFIS, or the specific system used by the employer

Skill: The operational software platform that the load controller uses to generate every load sheet

Why the field cares: Load control proficiency is inseparable from fluency with the departure control system used in the operation. Altea DCM (part of the Amadeus Altea suite) is the most widely used DCS at major European and international network carriers; SITA UFIS is used at major hubs by several airlines; Kestrel and Radixx serve specific carrier communities. The specific workflow — how the passenger data flows from the check-in module to the load control module, how the cargo manifest is entered or imported, how the fuel figure is applied, how the trim sheet is generated and formatted for ACARS transmission, and what the procedure is when the system flags an exception — differs between systems but is learned quickly by candidates who understand the underlying weight and balance principles. Candidates who have used the specific system used by the employer they are applying to, and who can demonstrate this by discussing specific workflow steps, are immediately differentiated.

Done signal: You are fluent in at least one departure control system load control module — either Altea DCM, UFIS, or another system used in your target operation. You can describe the end-to-end workflow from receiving the final passenger count through generating and transmitting the load sheet. You know the specific exception messages the system generates for CG out of limits, weight over limits, and LMC requirements, and you know the procedure for each. If you do not yet have access to the specific system used by your target employer, you have identified it and pursued any available training materials or demo access.

3–4 hours

11. Understand the NOTOC — content, format, captain notification requirement, and what to do when a NOTOC is missing or incomplete

Skill: The dangerous goods notification document that the load controller is responsible for transmitting to the captain

Why the field cares: The Notification to Captain (NOTOC) is the document that informs the flight crew of the dangerous goods on board their aircraft, their UN numbers, class or division, quantities, and hold positions. IATA regulations require the captain to be informed of all dangerous goods on board before departure, and the load controller is typically responsible for ensuring this happens — either by physically delivering a paper NOTOC to the cockpit, by ACARS transmission to the aircraft’s multifunction display, or by ensuring the NOTOC is included in the captain’s pre-departure documentation package. The procedure when a NOTOC is missing — when cargo has been loaded that should have a NOTOC but doesn’t — is specific and procedural: the departure must be stopped, the cargo acceptance team must be contacted, and the dangerous goods classification must be confirmed before departure can be authorised. Knowing this procedure, and knowing what “going without a NOTOC” for a known dangerous goods shipment means for the captain’s legal exposure, is the DGR operational knowledge that every qualified load controller must have.

Done signal: You can describe what a NOTOC must contain — the UN number, proper shipping name, class or division, quantity, and hold position of each dangerous goods item — and explain

why each piece of information is required for the captain's situational awareness. You can describe the procedure when a DGR shipment is in the hold but no NOTOC exists, and explain what the load controller's obligation is in this situation. You can describe the segregation requirements that the load controller must verify from the NOTOC — specifically which dangerous goods classes must be separated from passengers, from foodstuffs, and from each other.

3–4 hours

12. Understand passenger count reconciliation — the final boarding check and what the load controller does when the numbers do not match

2–3 hours

Skill: The most common source of load sheet errors and the procedure that catches them

Why the field cares: The passenger count on the load sheet must match the number of passengers actually on board the aircraft when the door closes. The final boarding check — comparing the total passengers in the departure control system against the gate's final boarding count, the aircraft door count, and the passenger manifest — is the last verification before the load sheet is transmitted as final. When the numbers disagree — when the DCS shows 175 passengers but the gate shows 173 — the load controller must stop, reconcile the discrepancy, and not transmit the final load sheet until the count is confirmed. The most common cause of discrepancies is a no-show passenger whose baggage has been offloaded but whose passenger record has not been cleared from the departure, or a passenger who boarded through a different gate and was not scanned. Every load controller has encountered this situation, and the interview specifically probes what their reconciliation procedure is.

Done signal: You can describe the passenger count reconciliation procedure: where the final passenger count comes from (the gate agent's boarding scan count, the aircraft door count from the purser), how it is compared against the departure control system count, what the procedure is when they disagree (identify the discrepancy's source — missed boarding scan, aircraft transit passenger, DCS record error), who must be notified (gate agent, airline operations, captain in some cases), and what the consequence of not resolving the discrepancy is (the load sheet cannot be transmitted as final until the count is confirmed). You can describe how a no-show passenger's baggage affects the weight and balance if it has not been offloaded.

13. Prepare for a timed manual load sheet exercise — weight calculation, CG check, and limit verification under time pressure

4–6 hours

Skill: The technical test that many load control employers include in their selection or type assessment process

Why the field cares: Load control selection processes and type qualification assessments frequently include a timed manual load sheet exercise — a simplified scenario where the candidate must compute the take-off weight, zero fuel weight, CG index, and verify compliance with all structural and CG limits, within a specified time limit (typically 15–20 minutes). The exercise is not conceptually difficult, but the combination of precision requirement (kilograms, not rounded to hundreds), time pressure, and the need to simultaneously check multiple limits (MSTOW, MSZFW, CG forward limit, CG aft limit, hold weight limits) creates a performance challenge that benefits from specific practice. Candidates who have practised this type of timed exercise on the aircraft type relevant to the employer will perform substantially better than those encountering it for the first time.

Done signal: You can complete a timed manual weight and balance exercise for your target aircraft type — given a dry operating weight, a fuel weight with its index contribution, passenger counts and cabin zone distributions, cargo weights with their hold positions and index values — within 20 minutes, arriving at the correct take-off weight, zero fuel weight, and CG index, and correctly identifying whether each result is within or outside the applicable limit. You have practised this at least three times for the specific aircraft type the employer operates.

SENIOR LOAD CONTROLLER / SUPERVISOR

All certified controller tasks still apply. Operational stories must demonstrate independent authority — situations where the senior controller made a call, not just followed a procedure.

14. Prepare to walk through the most technically complex or unusual loading situation you have personally managed 2–3 hours

Skill: Applied load control expertise in a non-standard scenario

Why the field cares: Senior load control interviews are anchored in the candidate's experience with genuinely unusual loading situations — not the standard daily operations, but the situations where the standard procedure did not have a clean answer: an exceptionally heavy individual item that needed specific floor load analysis before loading authorisation, an outsize cargo requiring a modified hold configuration, a large group of passenger weight deviation, or a cargo substitution that shifted the CG close to the aft limit at a weight where the performance team needed to be consulted. The story reveals whether the candidate's response was procedurally complete (check all limits, consult the relevant authority, document the authorisation, update the load sheet) or whether corners were cut under time pressure.

Done signal: For one loading situation outside the routine that you personally managed — not escalated, but managed — you can describe the specific scenario (the unusual cargo, the weight or CG challenge, or the operational complexity), the analysis you performed (the specific weight and CG calculation, the floor load check, or the limit comparison), the authorities you consulted if required (airline technical, performance team, aircraft manufacturer technical support), the authorisation you obtained, the documentation you created, and the outcome. The story demonstrates that you knew when a situation was beyond standard procedure and what the appropriate escalation path was.

15. Understand quality audit procedures for load control — proficiency check content, common errors, and the audit documentation 3–4 hours

Skill: The quality assurance framework that maintains load control accuracy across an operation

Why the field cares: Load control proficiency is maintained through mandatory recurrent training and proficiency checks — typically every 12 or 24 months depending on the airline's approved training programme. The proficiency check typically covers a manual weight and balance calculation, a dangerous goods scenario, a LMC procedure check, and a review of recent significant errors or incidents in the operation. Load control supervisors and trainers who design and administer these checks must understand what the most common error types are in their operation, how to design a check scenario that tests the competencies most likely to fail under operational pressure, and how to document the check result in a way that meets the airline's ATO (Approved Training Organisation) requirements. The most common load control errors — as documented in safety reports and ECCAIRS data — are: incorrect passenger count resulting from reconciliation failure, incorrect fuel figure from a late uplift change, incorrect LMC application resulting in an unrecalculated CG change, and hold loading exceedances from cargo weight misrepresentation.

Done signal: You can describe the content of the load control proficiency check required by your operation — what scenarios are used, what competencies are assessed, what the pass/fail criteria are, and how the result is documented. You can name the three most common load control error types in commercial aviation based on safety report data, explain why each type occurs (the operational pressure point that leads to the error), and describe what the quality audit approach does to reduce each type. You can describe the procedure when a load controller fails their proficiency check.

16. Prepare to lead a discussion on a CG management challenge — an aircraft loaded to the aft CG limit and the operational decisions it requires 3–4 hours

Skill: CG management authority at the level of a senior controller or supervisor

Why the field cares: The aft CG limit is the most safety-critical CG boundary — an aircraft loaded past its aft CG limit will have reduced pitch stability, may experience pitch-up during takeoff, and

3–4 hours

may be uncontrollable in some configurations. Approaching the aft limit requires specific management: the load controller may need to redistribute cargo or passenger seats to move the CG forward, consult the performance team for an assessment of the aft limit margin, notify the captain of the aft loading, and ensure that any LMC affecting the aft CG is managed with specific authority from the airline's operations team. The discussion that most reveals senior CG management competence is the one where the candidate describes a specific aft-CG situation they managed — the specific aircraft, the specific CG position relative to the limit, the actions taken to move the CG forward, and the specific authority obtained before departure.

Done signal: You can describe the operational actions available to a load controller when an aircraft is loaded near the aft CG limit: passenger seat reassignment (which seat zones can be reassigned to move CG forward, and by how much per seat zone), cargo redistribution between holds, fuel loading optimisation (where applicable), and captain notification and briefing. You can explain what the performance implication of aft CG is for takeoff — why the pilot must apply more down elevator to rotate, and why this reduces the effective performance margin. You can describe the specific authority chain for approving departure at a CG that is within limits but in the aft portion of the envelope.

3–4 hours

17. Understand the airline's approved training organisation (ATO) framework for load control — how load control training is approved and maintained

Skill: The regulatory training framework that governs load control certification

Why the field cares: Load control is not a licensed profession in the same way as the FAA A&P or EASA Part-66 — there is no individual regulatory certificate equivalent to the AME licence. However, airlines and ground handlers who perform load control under EASA regulations are required to have an Approved Training Organisation (ATO) approval for load control training, covering the initial qualification course, the aircraft type endorsement process, and the recurrent training and proficiency check programme. Understanding how the ATO framework applies to load control — what the training organisation must document, what the type endorsement requires, and what the recurrent training interval is — is the regulatory training literacy that trainers and supervisors in load control must have. IATA's Standard Ground Handling Agreement (SGHA) sets the baseline for ground handler training requirements; individual airline operations specifications (OpSpecs) may impose additional requirements.

Done signal: You can describe the regulatory framework governing load control training at your operation — whether it operates under an EASA ATO, under IATA SGHA requirements, under airline-specific operational specifications, or under a combination. You can describe what the initial load control qualification course must cover, what the aircraft type endorsement requires (the type-specific WBM, Aircraft Loading Manual, and load system training, plus a qualification assessment), and what the recurrent training interval and proficiency check requirement are. You can explain what the load control supervisor's or trainer's obligation is when a load controller's type endorsement lapses.

3–4 hours

18. Build quantitative estimation fluency for load control — CG shifts from common changes, weight impacts of standard passenger changes

Skill: Senior-speed reasoning about load sheet implications without running the system

Why the field cares: Senior load controllers are expected to give the operations team an immediate preliminary assessment of whether a proposed change is likely to be manageable within limits or will require significant rework — before running the system calculation. "If we change three passengers from economy rows 15–20 to business class, what direction does the CG move, and by roughly how much?" or "if we remove 500 kg of cargo from hold 1 forward because it didn't arrive, what happens to the CG?" The ability to make these estimates quickly — from the index arm values of the relevant zones, which every experienced load controller knows for their primary aircraft type — allows the operations team to make faster decisions and allows the senior controller to flag likely limit violations

before they are discovered in the system calculation.

3–4 hours

Done signal: You can estimate, for the aircraft type you are most experienced on, the approximate CG shift in %MAC for the following changes: moving 5 passengers from economy rear to economy forward; removing 500 kg of cargo from the forward hold; moving 3 passengers from economy to business class; adding 1,000 kg of fuel in the wing tanks versus the centre tank. Your estimates are approximate — correct in direction, and within $\pm 0.5\%$ MAC for the magnitude — and you explicitly state that they are estimates to be confirmed in the load system before any departure decision is made.

ALL LEVELS

19. Prepare a specific answer to "Why load control — and why this employer?" that reflects genuine operational engagement

1–2 hours

Skill: Technically specific motivation that distinguishes candidates who have thought about the role from those who have applied broadly

Why the field cares: Load control employers — particularly supervisors and quality managers conducting the interview — are looking for candidates who understand what the role actually involves and why it interests them specifically. The answer that says "I want to work in aviation operations" without assigning the specific appeal of load control — the combination of numerical accuracy, aviation technical knowledge, and the direct safety consequence of the work — is not sufficient. The expected answer in 2025 engages with one of: the direct safety accountability of the load sheet and the specific appeal of working in a role where accuracy has immediate consequence; the technical interest in the weight and balance problem itself — the way that fuel burn, passenger distribution, and cargo configuration interact to determine whether an aircraft is safely loaded; the operational pace and decision-making environment of a busy hub departure bank; or the specific operation at the target employer and what fleet or route mix makes it operationally interesting. The answer must be specific enough that it could not be given unchanged to a candidate applying for a check-in agent role or an airside ground crew role.

Done signal: You have a 90-second answer that names the specific aspect of load control that attracts you (the safety accountability, the technical weight and balance problem, the operational pace, or the specific employer's operation), connects it to something specific about your background (numerical work in a previous role, existing knowledge of a specific aircraft type, interest in the dangerous goods aspect), and explains why this employer's operation — their fleet, their handling contracts, their training programme, or their hub location — is specifically interesting to you. You have researched the employer beyond their website homepage.

20. Map your type qualifications, system experience, and operational background to a specific employer and role — and understand the career progression

1–2 hours

Skill: Self-positioning in a field where type qualifications and system familiarity matter as much as years of experience

Why the field cares: The load control job market is structured around type qualifications (which aircraft types you are endorsed on) and system familiarity (which departure control system you have operational experience with). A load controller qualified on A320 and B737 who has Altea DCM experience is immediately employable at a large fraction of the European network carrier ground handling contracts. A load controller qualified only on ATR-72 who has used only a carrier-proprietary system is in a narrower market. The career progression from load controller to senior load controller to duty manager or load control trainer follows a predictable structure, but the branches that lead to airline performance engineering or ground operations management require additional study — a part-time ATC degree, a professional qualification in aviation management, or internal development programme engagement. Knowing which branch you are targeting and what the specific next steps are — not vaguely "I'd like to progress into management" but "I am targeting the duty manager

1–2 hours

pathway and the next step is the shift supervisor role, which I understand requires 3 years of senior controller experience at this operation plus a management certificate” — is the career self-awareness that impresses interviewers at the supervisory level.

Done signal: You can state in under 60 seconds: the aircraft types you are qualified on or are training toward, the departure control system you have operational experience with (or are actively learning), the type of operation (short-haul, long-haul, cargo) where most of your experience has been gained, and two specific technical strengths you bring to the target role. You can describe the next one or two steps in your intended career progression and what is required to achieve them at the target employer.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Understand the load sheet — what it authorises, what an error means, one accident reference	All	3–4 h
2	Manual CG calculation using the index method — practised on a real aircraft type	All	6–8 h
3	Maximum structural weights — MTOW, MZFW, MLW, what limits each	All	3–4 h
4	STAR stories — accuracy under pressure, error disclosure, communication, asking for help	All	2–3 h
5	MEL interface for load control — how MEL items create loading restrictions	All	2–3 h
6	IATA DGR Categories 6 and 8 — load controller obligations, NOTOC process	All	4–5 h
7	Last Minute Change procedure — what triggers it, tolerance, documentation, captain notification	Certified+	3–4 h
8	Hold loading — per-bay limits, floor load calculation, bulk cargo distribution	Certified+	3–4 h
9	Load sheet discrepancy story — found it, fixed it, documented it, caused a delay	Certified+	2 h
10	DCS operation — Altea, UFIS, or employer-specific system end-to-end workflow	Certified+	4–6 h
11	NOTOC — content, captain notification, missing NOTOC procedure	Certified+	3–4 h
12	Passenger count reconciliation — final boarding check, discrepancy procedure	Certified+	2–3 h
13	Timed manual load sheet exercise — weight calculation, CG check, limits under time pressure	Certified+	4–6 h
14	Complex loading situation story — non-standard scenario, analysis, authority, documentation	Senior	2–3 h
15	Quality audit procedures — proficiency check content, common errors, ATO documentation	Senior	3–4 h
16	Aft CG management discussion — redistribution options, performance implication, authority	Senior	3–4 h
17	ATO framework for load control — training approval, type endorsement, recurrent interval	Senior	3–4 h
18	Quantitative CG estimation — direction and approximate magnitude for common changes	Senior	3–4 h
19	"Why load control — why this employer?" — technically specific	All	1–2 h
20	Type qualifications / system experience / career progression self-positioning	All	1–2 h

THE STRUCTURAL REALITY THAT MAKES AIRCRAFT LOAD CONTROL DIFFERENT

The structural reality that makes aircraft load control different from every other airport operations role: Aircraft load control is the only airport operations role where a single document issued by one person authorises a specific weight, a specific weight distribution, and a specific centre of gravity position for a flight carrying hundreds of people — and where that person's name, signature, or system credentials are permanently attached to that document in the aircraft's departure record.

This is not a dramatic framing. It is a regulatory and legal fact. The load sheet is the captain's authority to depart at the stated weight and CG. If the load sheet is wrong and the aircraft departs with incorrect weight or balance data, the captain is operating the aircraft with false information — and the load controller who signed that sheet is the person whose judgement or accuracy was deficient. There are accident reports with exactly this chain of events in the record.

Understanding this accountability — not as a source of anxiety but as the specific professional obligation that gives the role its character — is the beginning of every successful load control career. The load controller who understands their accountability processes every departure with a specific question running in the background: "am I certain enough of these numbers to sign this document?" When the answer is yes, the sheet is signed and transmitted. When the answer is "I need to check one more thing," the aircraft waits.

The career pathway that begins at load controller and ends at airline performance engineering, ground operations management, or airline operations control is not automatic — it requires additional study, additional qualification, and the professional curiosity to understand not just how to produce the load sheet but why every number on it is what it is and what would happen if it were different. But that pathway is open, it is well-documented, and it begins with the discipline to sign only what you can certify is correct. **The roles that are genuinely satisfying in load control are those where this accountability is combined with the operational pace and variety that makes aviation ground operations distinctively different from any other workplace. On a morning departure bank at a busy European hub, the load controller is simultaneously managing thirty aircraft, each with its own passenger count, cargo manifest, fuel figure, NOTOC, and CG — all of it moving, all of it subject to last-minute change, and all of it requiring a signed document before the aircraft can push back. The combination of mathematical accuracy, aviation technical knowledge, operational judgment, and safety accountability — all at high tempo — is what makes the role genuinely demanding and genuinely engaging.**

The career pathway that begins at load controller and ends at airline performance engineering, ground operations management, or airline operations control is not automatic — it requires additional study, additional qualification, and the professional curiosity to understand not just how to produce the load sheet but why every number on it is what it is and what would happen if it were different. But that pathway is open, it is well-documented, and it begins with the discipline to sign only what you can certify is correct.

BREAK INTO AEROSPACE
